



CLIENT DELIVERABLE

SAOS Service Manual

Systack Documentation

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ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b

ROLE	COLOR	HEX
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

SAOS Service Manual

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What's New in v7.0

- **Live Operations** endpoint — Real-time agent heartbeat monitoring
- **Activity** endpoint — Full audit trail of task lifecycle events
- **Dynamic Services** — Services fetched from backend config, not hardcoded
- **Deliverable Storage** — Upload/download/list endpoints for file management
- **Stripe Integration** — Tier-specific checkout links and pricing

Overview

The SAOS Customer Portal v2.1 is a single-page web application that gives clients real-time access to their AI agent fleet. This manual covers all system components, authentication, and operational details.

System Architecture

```
Client Browser
  ↓ (HTTPS via Tailscale)
Tailscale Serve (reverse proxy /dashboard/)
  ↓
Flask API (api.py, port 8768)
  ├── Static file serving (index.html)
  ├── Auth endpoints (/api/auth/*)
  ├── Portal endpoints (/api/portal/*)
  ├── Chat endpoints (/api/chat/*)
  ├── Deliverable endpoints (/api/deliverables/*)
  ├── PDF downloads (/download/*)
  └── PostgreSQL (systack_memory DB)
```

Authentication

PIN-Based Authentication Flow

1. Admin generates **temporary PIN** (6 digits, 24-hour expiry)
2. Client registers with temp PIN → sets **permanent PIN** (4–10 digits)

- 3. Client logs in with Client ID + PIN
- 4. System returns **Bearer token** (SHA-256 hashed, 30-day expiry)
- 5. Token stored in browser localStorage
- 6. All API calls include: `Authorization: Bearer <token>`

Auth Endpoints

ENDPOINT	METHOD	AUTH REQUIRED	PURPOSE
/api/auth/login	POST	No	Login with Client ID + PIN
/api/auth/register	POST	No	First-time PIN setup from temp PIN
/api/auth/forgot-pin	POST	No	Request PIN reset
/api/auth/change-pin	POST	Bearer	Change PIN while logged in
/api/auth/logout	POST	Bearer	Revoke current token
/api/auth/me	GET	Bearer	Get current client info
/api/portal/onboarding-status/{id}	GET	No	Check if client needs setup

Dashboard Tabs

1. Chat Tab

- Conversation sidebar with unread badges
- Real-time messaging with AI agents
- Auto-task creation on action keywords (“create”, “build”, “fix”, “deploy”, “update”)
- System messages for task creation confirmation
- Typing indicator
- 5-second polling for updates

2. Dashboard Tab

- **Metrics:** Active, Pending, Completed tasks, Unread messages
- **Agent Fleet:** Clickable cards with avatar, role, description, capabilities, status
- **Status colors:** IDLE (gray), BUSY (cyan), ERROR (red), OFFLINE (gray)

3. Services Tab

- Tier-specific service cards (active vs. pending)
- Infrastructure details (VPS specs, AI models, network)
- Support information (channel, response time, SLA)
- Current plan badge display
- **Dynamic loading:** Services fetched from /api/portal/services based on client tier

4. Tasks Tab

- Full task list with filter buttons
- **Filters:** All, Pending, Running, Done, Failed
- **Columns:** Task name, description, agent, status, created
- **Click any row** → opens detail modal with full info + error messages
- **File upload:** Paperclip icon in task detail → secure storage
- **File download:** Download links for agent-delivered files

5. Activity Tab

- Real audit trail of task lifecycle events
- Status icons (🔄 Running, ⌚ Pending, ✅ Done, ❌ Failed, 💀 Dead, 📁 Upload)
- **Clickable items** → same detail modal as Tasks tab
- Uses display_name + description

6. Live Operations Tab

- Real-time agent heartbeat monitoring
- 💖 **Active:** Agent responded within last 5 minutes
- ⚠️ **Stale:** No response in 5+ minutes
- 🛑 **Offline:** Agent not responding
- Shows last heartbeat timestamp per agent

7. Docs Tab

- PDF downloads with tier-specific filtering
- Documents: Quick Start, User Guide, Service Manual, Architecture, Mobile Guide

8. Settings Tab

- **Account Info:** Client ID, name, email, plan badge, status
- **Security:** Change PIN (requires current PIN)
- **Deployment Info:** Type, VPS status, Tailscale URL

API Endpoints (v2.1)

ENDPOINT	METHOD	AUTH	PURPOSE
/api/portal/health	GET	None	Health check
/api/portal/onboarding-status/<id>	GET	None	Check if client needs PIN setup
/api/auth/login	POST	None	Login with Client ID + PIN
/api/auth/register	POST	None	First-time PIN setup from temp PIN
/api/auth/forgot-pin	POST	None	Request PIN reset via

ENDPOINT	METHOD	AUTH	PURPOSE
			email
/api/auth/change-pin	POST	Bearer	Change PIN (logged-in clients)
/api/auth/logout	POST	Bearer	Revoke current token
/api/auth/me	GET	Bearer	Get current client info
/api/portal/status	GET	Bearer	Fleet overview, metrics, agents
/api/portal/tasks	GET	Bearer	Task list with descriptions
/api/portal/activity	GET	Bearer	Real activity audit trail
/api/portal/services	GET	Bearer	Tier-specific services (dynamic)
/api/portal/agents	GET	None	Agent list with roles + descriptions
/api/portal/client	GET	Bearer	Client info
/api/chat/conversations	GET	Bearer	List chat conversations
/api/chat/conversations	POST	Bearer	Create new conversation
/api/chat/conversations/<id>/messages	GET	Bearer	Get messages
/api/chat/conversations/<id>/messages	POST	Bearer	Send message (auto-creates task)
/api/chat/conversations/<id>/close	POST	Bearer	Close conversation
/api/chat/webhook/agent-response	POST	None	Agent → chat (n8n/OpenClaw webhook)
/api/chat/poll	GET	Bearer	Poll for new messages + task updates
/api/deliverables/upload	POST	Bearer	Upload file deliverable
/api/deliverables/list	GET	Bearer	List client deliverables
/api/deliverables/download/<filename>	GET	Bearer	Download file deliverable
/download/*	GET	None	PDF document downloads

Database Schema

saos_clients (enhanced v2.1)

COLUMN	TYPE	DESCRIPTION
id	INTEGER	Primary key
customer_name	VARCHAR	Display name
customer_email	VARCHAR	Registered email
tier	VARCHAR	Plan tier
auth_pin	VARCHAR(10)	Permanent PIN
temp_pin	VARCHAR(10)	Temporary PIN (24hr)
temp_pin_expires_at	TIMESTAMP	Expiry for temp PIN
onboarding_status	VARCHAR	pending / pin_set / active / suspended
last_login_at	TIMESTAMP	Last login timestamp
login_count	INTEGER	Total login count

task_queue (enhanced v2.1)

COLUMN	TYPE	DESCRIPTION
id	SERIAL	Primary key
task_type	VARCHAR	Machine type
display_name	VARCHAR	Human-readable name
description	TEXT	Full description
status	VARCHAR	PENDING / RUNNING / DONE / FAILED / DEAD
assigned_agent	VARCHAR	Agent responsible
priority	INTEGER	1-10, 5 default
payload_json	JSONB	Request details + deliverable info

agent_state (enhanced v2.1)

COLUMN	TYPE	DESCRIPTION
agent_name	VARCHAR	Primary key
role	VARCHAR	Short title
role_description	TEXT	Full explanation
capabilities	JSONB	Array of skill strings
avatar_emoji	VARCHAR(10)	Visual icon
tier_access	VARCHAR	all / specific tier
status	VARCHAR	IDLE / BUSY / ERROR / OFFLINE
last_heartbeat	TIMESTAMP	Last activity timestamp

client_auth_tokens

COLUMN	TYPE	DESCRIPTION
id	SERIAL	Primary key
client_id	INTEGER	Foreign key
token_hash	VARCHAR(64)	SHA-256 hash
expires_at	TIMESTAMP	Token expiry
last_used_at	TIMESTAMP	Last activity
revoked_at	TIMESTAMP	Revocation time

chat_conversations

COLUMN	TYPE	DESCRIPTION
id	SERIAL	Primary key
client_id	INTEGER	Owner
title	VARCHAR	Conversation title
status	VARCHAR	active / closed / archived
created_at	TIMESTAMP	
updated_at	TIMESTAMP	
closed_at	TIMESTAMP	

chat_messages

COLUMN	TYPE	DESCRIPTION
id	SERIAL	Primary key
conversation_id	INTEGER	Parent thread
sender_type	VARCHAR	client / agent / system
sender_name	VARCHAR	Display name
sender_agent	VARCHAR	Agent ID (if agent)
content	TEXT	Message body
message_type	VARCHAR	text / task_created / task_update / system
task_id	INTEGER	Linked task
read_at	TIMESTAMP	Read receipt
created_at	TIMESTAMP	

admin_client.py

COMMAND	PURPOSE
<code>--create-temp-pin {id}</code>	Generate temporary PIN for onboarding
<code>--list-clients</code>	Overview of all clients
<code>--client-status {id}</code>	Detailed client info

Example Usage

```
# Generate temp PIN
python3 admin_client.py --create-temp-pin 1

# List all clients
python3 admin_client.py --list-clients

# Check specific client
python3 admin_client.py --client-status 1
```

Service Tiers

TIER	PRICE	TARGET	KEY FEATURES
Personal	\$49/mo	Individuals	Email triage, calendar, reminders, research
Personal+	\$99/mo	Power users	+ Voice, multi-device, expense tracking
Business	\$299/mo	SMBs	Invoice processing, lead qualification, team chat
Enterprise	\$799/mo	Large orgs	On-premise, HIPAA, white-glove, 4hr SLA
Private	\$799/mo	Regulated	Air-gapped, zero cloud, audit trail
Accelerate	\$249/mo	Startups	Cloud GPU, auto-scaling, fast deployment

Stripe Checkout Links

Each tier has associated Stripe checkout URLs for monthly and annual billing.

Security

Network Security

- **Tailscale WireGuard** encryption for all traffic
- Tailnet-only access (no public internet exposure)
- HTTPS via Tailscale TLS termination

Application Security

- PIN-based authentication (4–10 digits)

- Session tokens hashed with SHA-256
- 30-day token expiry with revocation
- Client-scoped data access (all queries filter by client_id)
- No cross-client data leakage

Data Security

- PostgreSQL stored locally on VPS
- No cloud data exposure (except Stripe — client’s own account)
- All AI processing on local Ollama models
- No third-party AI API calls with client data

Troubleshooting

SYMPTOM	CAUSE	SOLUTION
“Invalid PIN”	Wrong PIN	Try again or use Forgot PIN flow
“PIN not set”	No auth_pin in DB	Generate temp PIN for client first
“Temp PIN expired”	>24 hours	Generate new temp PIN
“Client not found”	Wrong ID	Verify numeric Client ID
“Session expired”	Token expired	Re-login with Client ID + PIN
Page blank on mobile	Tailscale not connected	Connect Tailscale VPN
PDF won’t download	Wrong URL path	Use <code>/dashboard/download/</code> prefix
“Service Temporarily Unavailable”	API restarting	Click Retry or wait 30s
Agent stale	Processing large task	Check Tasks tab, wait or contact support

Version History

VERSION	DATE	CHANGES
v1.0	2026-06-22	Initial dashboard (Overview, Agents, Tasks, Account)
v2.0	2026-06-24	Complete rebuild: Chat, Dashboard, Services, Tasks, Docs. Tier content. Token auth.
v2.1	2026-06-25	PIN onboarding, agent descriptions, task detail modals, Settings tab, 7 total tabs
v2.1+	2026-06-29	Live Ops tab, Activity audit trail, dynamic services, deliverable storage, error fallback